

A New Interactive Music System by Fusion of Melody Morphing and Body Movements

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Figure 1: Automatic performance with proposed system

ABSTRACT

The fusion of music-information processing and human physical functions will enable new musical experiences to be created. We developed an interactive music system, which provides users with the experience of conducting a musical performance. This system converts music arranged using melody morphing method based on the general theory of tonal music into electrical muscle stimulation to control the body movements of multiple performers (e.g., musicians and dancers) via devices attached to the performers' hands and feet. The melodies used in the system are divided into segments, and each segment has multiple variations of melodies. The user can interactively control the performers, thus the actual performance.

CCS CONCEPTS

• **Human-centered computing** → **Human-computer interaction (HCI)**.

KEYWORDS

Melody morphing; Musical performance; Electrical muscle stimulation (EMS)

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1 INTRODUCTION

New musical expressions can now be created due to developments in deep learning and the construction of physical interfaces for musical instruments. Attempts to make machines play music automatically began as far back as the 16th century, and there has been progress in automating the playing of instruments such as automatic pianos and percussion. In the 2010s, the invention of devices that can play multiple orchestral instruments simultaneously [8] and the conducting of sophisticated musical performances by robots [5] represent a new frontier in automated performance systems. As part of the research on augmented humans, there has been much effort in supporting musical-instrument performance by controlling the hand and finger movements of musicians using myoelectric stimulation and exoskeletal devices [10, 11]. Electrical muscle stimulation (EMS) uses electrical impulses to elicit action on the motor fibers/nerves, causing the involuntary contraction of the user's muscles. EMS, while well known as a tool for physiotherapy, offers the potential to remotely manipulate a performer's hand movements, modifying his/her performance [7, 9]. On the basis of these technological trends, we believe that the combination of the latest music-information processing technology and human physical functions will enable future high-quality musical expressions.

We propose an interactive music system called "Deep Augmented Performers," which enables melodies to be played by performers who are manipulated by a user. With Deep Augmented Performers, the user converts music arranged from melody morphing into EMS and controls the body movements of multiple performers, such as dancers and musicians, on the stage, via devices attached to

the performers' hands and feet. Variations of melodies are composed on the basis of the generative theory of tonal music (GTTM) [6], whereby switching the melody segments maintains the overall structure of the melody and only changes the ornamentation. In ensemble music performance, the conductor provides instructions on timing and performance expression, bringing unity to the performance. Musical performance naturally involves gestures, but the acquisition and understanding of such gestures requires a great deal of learning. Reflecting on human-computer symbiosis, Our aim is transferring skills to enable laypersons to carry out complex actions on the spot.

We believe that a computer that can deeply understand music must be developed to achieve a composition support system that enables various musical operations. Therefore, for the past 15 years, we have been researching and developing systems based on GTTM such as those proposed by Lerdahl and Jackendoff [2]. The performance of GTTM analysis via computers has dramatically improved by introducing deep learning [4]. The time-span tree, which is a deep structure obtained as a result of a GTTM analysis, structurally shows the relationship between the important notes and ornamentation notes. The main advantage of using the time-span tree is that it is possible to reduce the notes of a melody by decreasing the ornamentation notes step by step while maintaining the main structure of the melody. One example of a melody manipulation using the time-span tree is melody morphing, which generates an intermediate melody from two input melodies [3]. To the best of our knowledge, this is the first study on the fusion of music-information processing and human physical functions using EMS.

2 MELODY MORPHING

Deep Augmented Performers enables melodies to be manipulated using melody morphing. Figure 2 is an example of abstracting a melody by using a time-span tree. There is a time-span tree from melody D, which embodies the results of the GTTM analyses. The structurally important notes are connected to a branch nearer the root of the tree. In contrast, ornamentation notes are connected to the leaves of the tree. We can obtain an abstracted melody, melody E, by slicing the tree in the middle and omitting notes that are connected to branches under line E.

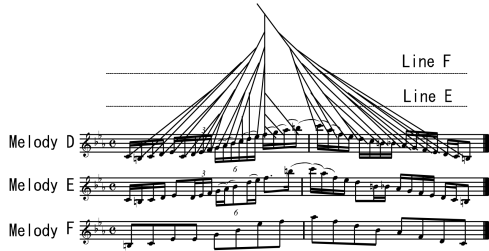


Figure 2: Time-span reduction of GTTM

In melody morphing, we use the primitive operations of the subsumption relation (written as $\sqsubseteq$), meet (written as $\sqcap$), and join (written as $\sqcup$). A subsumption represents the relation by which “an instantiated object” $\sqsubseteq$ “an abstract object.” The meet operator extracts the largest common part or the most common information of the time-span trees of two melodies in a top-down manner. The

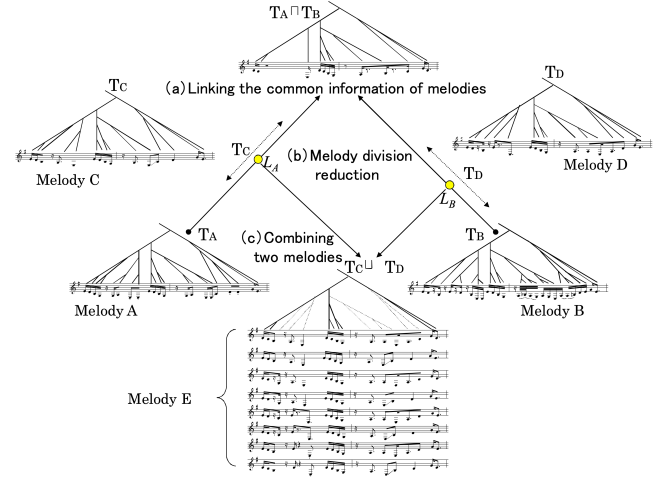


Figure 3: Melody morphing

join operator joins two time-span trees in a top-down manner as long as the structures of two time-span trees are consistent.

By using time-span trees T_A and T_B from melodies A and B, we can calculate the most common pieces of information, $T_A \sqcap T_B$, which are the essential parts of melody A as well as those of melody B. The meet operations $T_A \sqcap T_B$ are abstracted from T_A and T_B , and the discarded notes are regarded as the difference between T_A and T_B 's information (Figure 3a). We argue that there are features without the other melody that are contained in the difference information of T_A and T_B . Therefore, we need a method for smoothly increasing or decreasing these features. The melody-divisional-reduction method can abstract the notes of the melody in the differential branch of the time-span tree (Figure 3b). We use the join operator to combine melodies C and D, which are the results of the divisional reduction or augmentation, using the time-span tree of melodies A and B (Figure 3c).

3 EMS SYSTEM

Deep Augmented Performers uses an EMS system to establish control over simple movements, such as snapping wrist joints and flexing elbow joints, of multiple performers. The EMS system includes a device that generates electrical stimulation by using an Arduino circuit (an electrical stimulation circuit), personal computer, direct current stabilized power supply, and electrode pads, with reference to a previous study [1] (Figure 4). In the circuit, a 555 timer is used to generate pulses, and a transformer is used to increase the voltage. The stimulation pulse frequency is 40 to 70 Hz and the pulse width is 0.8 ms. The pulse voltage can be controlled by the user and varied depending on the type of muscle to be controlled. When the voltage of the DC-stabilized power source is increased, the electrical stimulus strengthens and the response of the human body is enhanced.

We conducted a feasibility study of playing several musical instruments using this EMS system (Figure 5) and confirmed that simple movements, such as snapping of the wrist joint and bending of the elbow joint, can be completely controlled. In other words, the bodies of the participant performers who received EMS were able to be moved independently of their own consciousness. We also

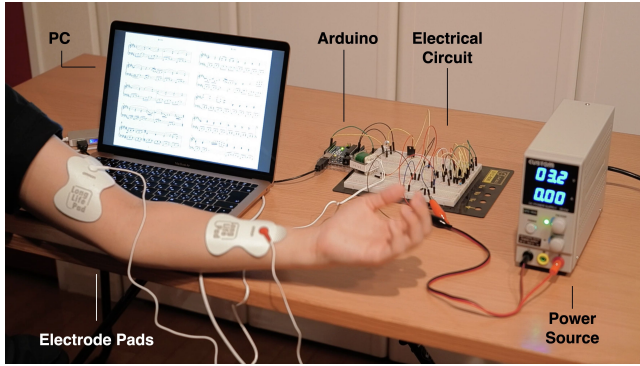


Figure 4: EMS system configuration



Figure 5: Feasibility study using EMS system

found that it was possible for a participant to perceive rhythm, such as from a metronome, by applying a weak electrical stimulus to the participant's back. However, it is still difficult with the system to control finer movements such as those involved with playing the guitar.

This EMS system can synchronize the body movements of many people by controlling multiple EMS generators in an integrated manner. This makes it possible to play various types of ensembles. We evaluated the time response of the EMS system by giving stimuli to ten participants (six men four women) at the same time for 20 times, while the participants were blindfolded, at a stimulation frequency of 60 Hz (Figure 6). The average delay in transmission was 242.8 ms (standard deviation 86.4 ms). The participants showed minor differences in their perception of the stimuli and magnitude of their movements, but these differences were almost eliminated through calibration.

4 DEEP AUGMENTED PERFORMERS

Deep Augmented Performers is ideal for the automatic playing of ensemble percussion instruments. It can convert music arranged from melody morphing into EMS and control the body movements of multiple performers, such as dancers and musicians, via devices attached to the performers' hands and feet. The melodies used in the system are divided into multiple segments, and each segment has multiple melody variations. We developed a graphical user interface (GUI) that enables anyone to easily manipulate performers; thus the actual performance (Figure 7). The operation method of the GUI is as follows. (a) By selecting two songs and pressing



Figure 6: Time-response evaluation by EMS system

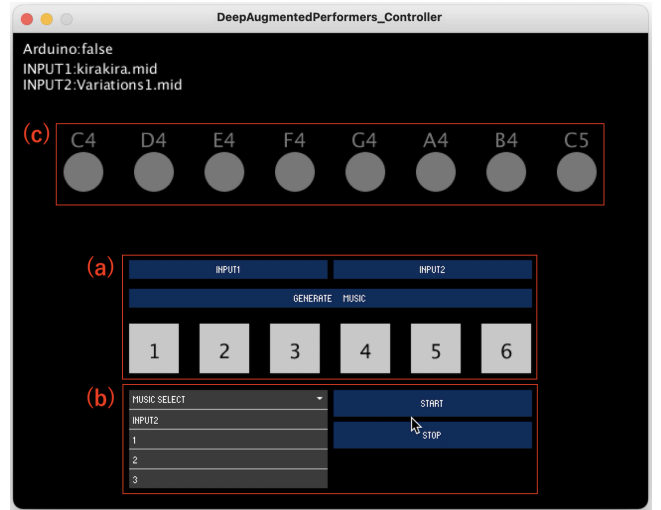


Figure 7: Proposed system's GUI

the GENERATE MUSIC button, multiple melodies obtained from melody morphing are generated (e.g., 1~6), (b) a song to play from the obtained melodies is selected, and (c) electrical stimulation is generated and sent to the electrodes corresponding to each sound (e.g., C4~C5). Therefore, the user can control performers by freely manipulating the melody.

We conducted a user survey on the effectiveness of Deep Augmented Performers that involved performances of four handbell players, one drummer, and one dancer (Figures 1 and 8). We determined the correct electrode-pad position to reproduce the movements of each performer by considering the movement of the muscles. The setting position and presentation method of the electric stimulus are as follows.

- Handbell players need to snap wrists when playing the bells. We set the location for stimulating the extensor carpi radialis longus and brevis, which extend the wrist joint. The handbell players could play the main melody generated from melody morphing in full control.
- For the drummer, we set the location for stimulating the bicep muscle of the upper arm, brachioradial muscle, and gastrocnemius muscle, which controls the plantar flexion of the foot joint. Because the forearm and wrist move, this would enable a drummer to play a beat with sticks. When



Figure 8: User study to evaluate Deep Augmented Performers

stimulating the gastrocnemius muscle with the heel on the floor and the foot pointed upward, it is possible to obtain rhythmic movement with the foot. The drummer could play the corresponding four beats with full control by reading the tempo information of the song being played.

- For the dancer, we applied a weak electrical stimulus to his back by reading the tempo information of the song being played. This enabled him to perceive rhythm, such as from a metronome, and dance any dance while keeping the rhythm in sync with the other participants.

The performance using this system lasted for about 5 minutes, and we confirmed that Deep Augmented Performers is effective by synchronizing the playing of the melody by using melody morphing and controlling performers' movements. However, we found that it is difficult to synchronize the timing perfectly with the EMS system due to slight timing delays. Therefore, we will attempt to solve this problem by attaching electrode pads at multiple locations to increase the reflection speed and synchronizing with the beat.

5 CONCLUSION

We proposed Deep Augmented Performers, which enables a user to manipulate music freely through melody morphing using the time-span tree of GTTM. The system controls the body movements of multiple performers to enable interactive performance. We will pursue a new style of musical experience by fusing music-information processing and human physical functions. The findings of this research have the potential to bring about changes in the relationship between humans and systems and performance experience. We hope that this paper will contribute to further developments in this field. We also plan to construct a system that enables more complex instrumental performances, such as piano and guitar. We will continue to diversify the parts and patterns to be controlled by

the EMS system and model methods of controlling more detailed operations.

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REFERENCES

- [1] Ayaka Ebisu, Satoshi Hashizume, and Yoichi Ochiai. 2018. Building a feedback loop between electrical stimulation and percussion learning. In *ACM SIGGRAPH 2018 Studio (SIGGRAPH '18)*. ACM, 1–2.
- [2] Masatoshi Hamanaka, Keiji Hirata, and Satoshi Tojo. 2007. Implementing 'A Generating Theory of Tonal Music'. *Journal of New Music Research* 35, 4 (2007), 249–277.
- [3] Masatoshi Hamanaka, Keiji Hirata, and Satoshi Tojo. 2008. Melody Morphing Method Based on GTTM. In *Proceedings of the 2008 International Computer Music Conference (ICMC '08)*, 155–158.
- [4] Masatoshi Hamanaka, Keiji Hirata, and Satoshi Tojo. 2017. deepGTTM-III: Multi-task Learning with Grouping and Metrical Structures. In *Proceedings of the 2017 International Symposium on Computer Music Multidisciplinary Research (CMMR '17)*, 238–251.
- [5] Yoichiro Kawaguchi. 2013. *Squarepusher × Z-MACHINES*. Retrieved February 24, 2022 from <https://youtu.be/VkUq4sO4LQM>
- [6] Fred Lerdahl and linguist Ray Jackendoff. 1983. *A Generative Theory of Tonal Music*. Cambridge, Massachusetts: MIT Press.
- [7] Pedro Lopes, Max Pfeiffer, Michael Rohs, and Patrick Baudisch. 2016. Hands-on Introduction to Interactive Electrical Muscle Stimulation. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '16)*. ACM, 944–947.
- [8] Pat Metheny. 2013. *Orchestrion - An Excerpt from The Orchestrion Project*. Retrieved February 24, 2022 from <https://youtu.be/evHVh4bqaOQ>
- [9] Jun Nishida and Kenji Suzuki. 2017. bioSync: A Paired Wearable Device for Blending Kinesthetic Experience. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '17)*. ACM, 3316–3327.
- [10] Nobuhiro Takahashi, Hayato Takahashi, and Hideki Koike. 2019. Soft Exoskeleton Glove Enabling Force Feedback for Human-Like Finger Posture Control with 20 Degrees of Freedom. In *Proceedings of IEEE World Haptics Conference (WHC '19)*. IEEE, 217–222.
- [11] Emi Tamaki, Miyaki Takashi, and Jun Rekimoto. 2011. Possessedhand: Techniques for controlling human hands using electrical muscles stimuli. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)*. ACM, 543–552.